

Q. 1. Suppose that A , B and C are three finite sets. For each of the following, determine whether or not it is true. Explain your answers.

(a) $(A - B = A) \rightarrow (B \subset A)$

(b) $(A \cap B \cap C) \subseteq (A \cup B)$

(c) $\overline{(A - B)} \cap (B - A) = B$

1. (a) $(A - B = A) \rightarrow (B \subset A) \equiv (A \cap B = \emptyset) \rightarrow (B \subset A)$

False. When B is $\emptyset$, $A \cap B = \emptyset$ and $B \subset A$, it's true.

When B is not $\emptyset$, A and B don't have a common element.

so $B \subset A$ is false. In conclusion, it is false.

(b) Suppose $x \in A \cap B \cap C$, then $x \in A$, then $x \in A \cup B$

True $\therefore (A \cap B \cap C) \subseteq (A \cup B)$

(c) **False.** Suppose $x \in A \cap B$, then $x \in B$. However, $x \notin B - A$,

thus $x \notin \overline{(A - B)} \cap (B - A) \therefore \overline{(A - B)} \cap (B - A) \neq B \therefore \text{False.}$

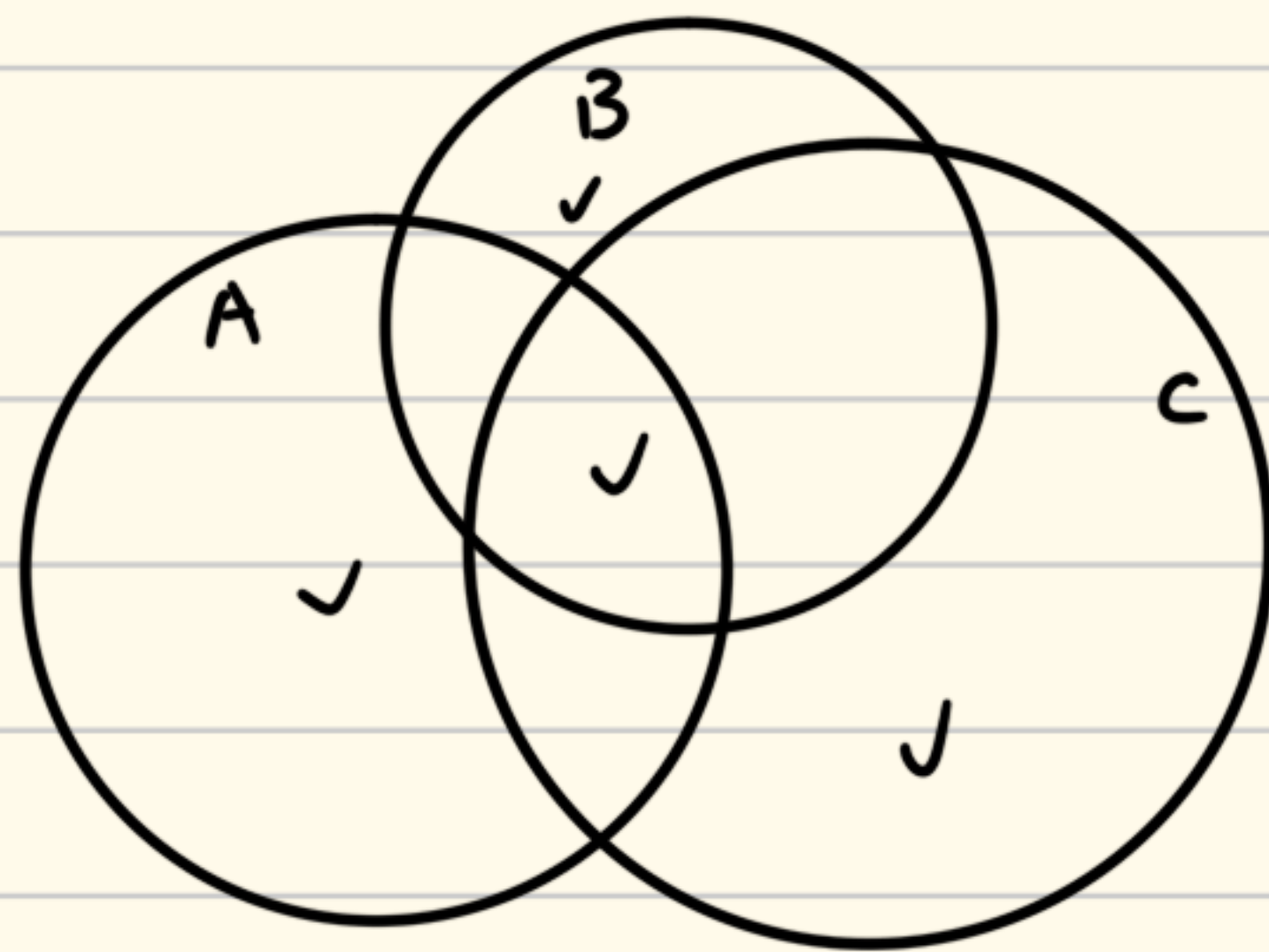
Q. 2. The symmetric difference of A and B , denoted by $A \oplus B$, is the set containing those elements in either A or B , but not in both A and B .

- (a) Determine whether the symmetric difference is associative; that is, if A , B and C are sets, does it follow that $A \oplus (B \oplus C) = (A \oplus B) \oplus C$?
- (b) Suppose that A , B and C are sets such that $A \oplus C = B \oplus C$. Must it be the case that $A = B$?

$$x \in A \wedge x \notin B \quad \left\{ \begin{array}{l} (1) x \in C \\ (2) x \notin C \Rightarrow x \notin A \cap C \Rightarrow x \in A \oplus C \end{array} \right.$$

Q. 3. Prove or disprove that there exists an infinite set A such that $|A| < |\mathbb{Z}^+|$.

2. (a)



A	B	C	$A \oplus (B \oplus C)$	$(A \oplus B) \oplus C$
1	1	1	1	1
1	1	0	0	0
1	0	1	0	0
1	0	0	1	1
0	1	1	0	0
0	1	0	1	1
0	0	1	1	1
0	0	0	0	0

From the membership table we can see that $A \oplus (B \oplus C) = (A \oplus B) \oplus C$

(b) Yes. Suppose $A \neq B$, let $x \in A \wedge x \notin B$.

then ① $x \in C$, $\therefore x \in A \cap C \therefore x \notin A \oplus C \therefore x \notin B \therefore x \in B \oplus C$.

which violates $A \oplus C = B \oplus C$.

② $x \notin C$, $\therefore x \in A \therefore x \in A \oplus C \therefore x \notin B, x \notin C \therefore x \notin B \oplus C$.

which violates $A \oplus C = B \oplus C$.

$\therefore A = B$.

3. False. Suppose it's true.

$\therefore |A| < |\mathbb{Z}^+| \therefore$ It's countable because we can list it

$\therefore A$ is finite or $|A| = |\mathbb{Z}^+| \therefore |A| < |\mathbb{Z}^+| \therefore A$ is finite,

which violates that A is infinite.

4.

Q. 4. Suppose that two functions $g : A \rightarrow B$ and $f : B \rightarrow C$ and $f \circ g$ denotes the composition function.

- (a) If $f \circ g$ is one-to-one and g is one-to-one, must f be one-to-one? Explain your answer.
- (b) If $f \circ g$ is one-to-one and f is one-to-one, must g be one-to-one? Explain your answer.
- (c) If $f \circ g$ is one-to-one, must g be one-to-one? Explain your answer.
- (d) If $f \circ g$ is onto, must f be onto? Explain your answer.
- (e) If $f \circ g$ is onto, must g be onto? Explain your answer.

$$A \xrightarrow{g} B \xrightarrow{f} C$$

$$g(x_1) = g(x_2)$$

$$f(y_1) = f(y_2)$$

(a) **No**. Let $A = \{1\}$, $B = \{0, 1\}$, $C = \{1\}$

$g(1) = 1$, $f(1) = 1$, $f(0) = 1$. Then $f \circ g$ is one-to-one, g is one-to-one but f isn't one-to-one.

Yes

(b) If g isn't one-to-one, then $\exists x_1, x_2$, s.t.

$g(x_1) = g(x_2) \therefore f \circ g(x_1) = f \circ g(x_2)$, which violates that $f \circ g$ is one-to-one

(c) **Yes**. If g isn't one-to-one, then $\exists x_1, x_2$, s.t.

$g(x_1) = g(x_2) \therefore f \circ g(x_1) = f \circ g(x_2)$, which violates that $f \circ g$ is one-to-one

(d) **Yes**. If f is not onto, then $\exists x \in C$, s.t. $x \notin f(B)$.

$\because g(A) \subseteq B \therefore f(B) \supseteq f(g(A)) \therefore x \notin f(g(A))$, which means $f \circ g$ isn't onto

$\therefore f$ is onto.

(e) **No**. Let $A = \{1\}$, $B = \{0, 1\}$, $C = \{1\}$

$$g(1) = 1, f(1) = 1, f(0) = 1$$

Here g is not onto, but $f \circ g$ is onto.

Q. 5. Give an example of two uncountable sets A and B such that the difference $A - B$ is (a) finite, (b) countably infinite, (c) uncountable. Note: one example for each subquestion (a), (b), or (c).

Q. 6. If A is an uncountable set and B is a countable set, must $A - B$ be uncountable?

5. (a) $A = B = \{x \mid x \in \mathbb{R}\}$

(b) $A = \{x \mid x \in \mathbb{N} \text{ or } 0 \leq x \leq 1, x \in \mathbb{R}\}$ $B = \{x \mid 0 \leq x \leq 1, x \in \mathbb{R}\}$

(c) $A = \mathbb{R}$, $B = \{x \mid x \in (0, 1), x \in \mathbb{R}\}$

6. Yes. $\because A = (A - B) \cup (A \cap B)$ If $A - B$ is not uncountable, thus is countable. $\because B$ is countable $\therefore A \cap B$ is countable. $\therefore A = (A - B) \cup (A \cap B)$ is countable, which violates that A is an uncountable set.

Q. 7. Show that the set $\mathbf{Z}^+ \times \mathbf{Z}^+$ is countable by showing that the polynomial function $f : \mathbf{Z}^+ \times \mathbf{Z}^+ \rightarrow \mathbf{Z}^+$ with $f(m, n) = (m + n - 2)(m + n - 1)/2 + m$ is one-to-one and onto.

Q. 8. Assume that $|S|$ denotes the cardinality of the set S . Show that if $|A| = |B|$ and $|B| = |C|$, then $|A| = |C|$.

Proof:

7. Let $k = m + n - 1$, $\because n \geq 1 \quad \therefore m \leq k$

$$\therefore f(m, n) = \frac{k(k-1)}{2} + m = 1 + 2 + \dots + k - 1 + m$$

Divide the $\mathbb{Z}$ into $\left(\frac{k(k-1)}{2}, \frac{k(k+1)}{2}\right]$ ($k \in \mathbb{Z}^+$)

There are $\frac{k(k+1)}{2} - \frac{k(k-1)}{2} = k$ elements in each interval,

which can be represented by $\frac{k(k-1)}{2} + m$ ($1 \leq m \leq k$).

① one-to-one: Each element only falls in 1 interval,

so there is a specific k . In interval, it has a unique position.

$\therefore$ There is a unique m . $\therefore$ It's one-to-one.

②. With the description above, every integer can be represented by $\frac{k(k-1)}{2} + m$, thus is onto.

8. $\because |A| = |B|, |B| = |C| \therefore$ There exists one-to-one correspondences

$f: A \rightarrow B, g: B \rightarrow C$. Next we want to prove $g \circ f$ is bijection.

Proof: ①. One-to-one: Suppose $g \circ f(x_1) = g \circ f(x_2)$, $\because g$ is injection

$\therefore f(x_1) = f(x_2) \therefore f$ is injection $\therefore x_1 = x_2 \therefore g \circ f$ is injection

② onto: $\because g$ is surjection $\therefore \forall z \in C, \exists y \in B, \text{s.t. } g(y) = z$.

$\because f$ is surjection $\therefore \exists x \in A, \text{s.t. } f(x) = y \therefore \exists x \in A, y \in B, \text{s.t. } g(f(x)) = z$.

$\therefore g \circ f$ is surjection.

$\therefore g \circ f$ is bijection $\therefore |A| = |C|$.

Q. 9. Suppose that $f(x)$, $g(x)$ and $h(x)$ are functions such that $f(x)$ is $\Theta(g(x))$ and $g(x)$ is $\Theta(h(x))$. Show that $f(x)$ is $\Theta(h(x))$.

Q. 10. Suppose that $f_1(x)$ is $\Theta(g_1(x))$ and $f_2(x)$ is $\Theta(g_2(x))$. Prove or disprove that $f_1(x) - f_2(x)$ is $\Theta(g_1(x) - g_2(x))$.

$$9. \therefore f(x) = \Theta(g(x))$$

$$\therefore f(x) = O(g(x)) \quad \therefore \exists C_1, k_1, \text{ s.t. } \forall x > k_1, |f(x)| \leq C_1 |g(x)|$$

$$\therefore f(x) = \Omega(g(x)) \quad \therefore \exists C_2, k_2, \text{ s.t. } \forall x > k_2, |f(x)| \geq C_2 |g(x)|$$

$$\therefore g(x) = \Theta(h(x))$$

$$\therefore g(x) = O(h(x)) \quad \therefore \exists C_3, k_3, \text{ s.t. } \forall x > k_3, |g(x)| \leq C_3 |h(x)|$$

$$\therefore g(x) = \Omega(h(x)) \quad \therefore \exists C_4, K_4, \text{ s.t. } \forall x > K_4, |g(x)| \geq C_4 |h(x)|$$

$$\therefore \forall x > \max(K_1, K_3), |f(x)| \leq C |g(x)| \leq C_1 C_3 |h(x)|$$

$$\therefore K'_1 = \max(K_1, K_3), \quad C'_1 = C_1 C_3 \quad \therefore f(x) = O(h(x))$$

$$\therefore \forall x > \max(K_2, K_4), |f(x)| \geq C_2 |g(x)| \geq C_2 C_4 |h(x)|$$

$$\therefore K'_2 = \max(K_2, K_4), \quad C'_2 = C_2 C_4 \quad \therefore f(x) = \Omega(h(x))$$

$$\therefore f(x) = \Theta(h(x))$$

10. Disproof: A counterexample: $f_1(x) = f_2(x) = x$, $g_1(x) = 3x$, $g_2(x) = 2x$

$$f_1(x) - f_2(x) = 0, \quad g_1(x) - g_2(x) = x$$

$\therefore f_1(x) - f_2(x) = \Theta(g_1(x) - g_2(x))$ is clearly false.